



Quality Management of major product quality deviations

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§ Complete overhaul of the document

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§ Complete overhaul of the document

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1 Purpose

The objective of this procedure is to describe the management of major deviation relating to Product Quality within BYTP.

The purpose of this procedure is to

Ensure that major deviations related to the quality of the works are reported within BYTP (independently of the communication made to the client) and are the subject of attention from all the management concerned.

Ensure that these major deviations are analysed, their root causes identified, and corrective and/or preventive actions are implemented to prevent their recurrence at the project and company level.

This procedure does not cover any remedial actions (treatments) carried out in relation to the defects, nor any actions taken with respect to the Client.

This procedure complements the fault management procedure referenced MNG-PRO-010.

2 Scope

This procedure applies to all BYTP projects and construction sites, within France and internationally.

By default, it applies to all activities under BYTP's responsibility (including BYTP subcontractors), in some cases, specific projects or entity governance rules prevail.

Application to BYTP co-contractors is not imposed given the limit of liability that applies with these companies.

3 Reference documents

- EN ISO 9001 standard, v. 2015 [Quality Management System – requirements]
- EN ISO 9000 standard, v. 2015) [Quality Management System – Essential Principles and Vocabulary]
- Documentation fascicle FD ISO 10005 v. 2018 [Quality Management System – Guidelines for Quality Plans]
- Documentation fascicle FD ISO 10006 v. 2018 [Quality management – Guidelines for quality management in projects]
- MNG-PRO-010 [Fault Management Procedure]
- MNG-MANUEL-0013 [Quality Manual]

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4 Definitions and abbreviations

Defect in product quality: A defect is a proven deviation from one or more contractual or technical requirements relating to the work. A defect can be **observable** (cracks, quality of facings, etc.) or **not observable** (non-compliance with a specification for material, ...).

Major Deviation = Major Defect or Major System error

- **Major Defect:** This is a **proven defect** in product quality that has resulted or could have resulted in a direct cost of scrapping or repairing exceeding **€50,000**.
- **Major system error:** This is a proven fault from company's standards and procedures.

Corrections: Actions aimed at repairing/ correcting the defects or non-conformity (actions outside the scope of this procedure)

Corrective actions: Actions aimed at eliminating the cause of a defect or non-conformance to prevent it from reoccurring at the site level

Preventive actions: Actions aimed at eliminating the cause of a defect or non-conformance to prevent it from reoccurring on a BYTP scale

BYTP : Bouygues Travaux Publics

PQ : Product Quality

QE: Quality Environment

ExCo: Executive Committee

5 Responsibilities

- **The Product quality process Leader** is:
 - In collaboration with the Product Quality Manager, responsible for this procedure and its application. He ensures its circulation, implementation and updates if necessary.
 - Informed of major quality deviations.
 - Recipient of root cause analysis and participates as much as possible in presentation of major deviation to the ExCo.
- **The Project Director / Site Production Manager** is in charge of:
 - Identifying the major deviation (with the support of its production teams).
 - Reporting major deviation to the Product Quality Manager and BYTP QE Manager.
 - Organizing with the support of Project Quality Manager the analysis of major deviation.
 - Defining and implementing with the help of their team, the corrective actions with the aim of avoiding recurrences on the project
 - Proposing preventive actions to avoid reappearances at BYTP level, which will be examined during the presentation of major deviations to the ExCo.

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- **The Project Quality Manager is in charge of:**
 - Carrying out and formalizing the root cause analysis of each major deviation with the support of Project Director. He may request the support of headquarters team, particularly for initial analysis.
 - Keeping BYTP's Product Quality Manager and QE Manager informed of the findings of the analysis.
 - Ensuring that corrective actions are implemented in the project.
- **The Quality and Environmental Manager of the country entity (if applicable)**
 - May be consulted to determine if it is a major deviation.
 - Is informed of major deviations.
 - Participates, if necessary, in the analysis of the major deviations.
 - Is informed of the corrective actions to be implemented for the project.
- **BYTP Product Quality Manager**
 - Is informed of the major deviations.
 - Lists the impacted standards together with BYTP's QE Manager.
 - Participates, if necessary, in the analysis of the major deviation.
 - Organizes and participates in the meeting to present major deviation to ExCo.
 - Monitors the implementation of preventive actions.
 - Circulates information related to major deviation.
 - Maintains the register of major deviations and associated indicators up to date.
- **BYTP QE Manager :**
 - Is informed of the major deviations.
 - Ensures the implementation of the procedure at the site level.
 - Lists the impacted standards together with QP Manager.
 - Participates in the presentation of the major deviations to ExCo.
 - Ensures the implementation of preventive actions.

6 Management of Major deviations in product quality

Refer to Appendix 1

6.1 Identify a major quality deviation

Any defect or system error relating to the quality of a work and falling within the definition of a major deviation must be clearly identified.

The Project Director/ Site Production Manager via his team and/or the Business Network Leader is responsible for this identification.

The Project quality manager and BYTP's QP manager can provide assistance in the event that the identification of the major deviation is subject to discussion.

Note that the notion of major deviation is internal to BYTP. It is not directly related to the opening of a Non-conformance report issued to the Client.

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Beyond the strict definition of a major deviation, it is important to consider the following questions when a quality deviation occurs in comparison to contract specifications, standards, or procedures:

1. Did the deviation have, or could it have had, a significant impact on the cost or schedule?
2. Could the deviation potentially occur again on another BYTP project?

If the answer to both of these questions is yes, the current procedure is applicable.

6.2 Report a major deviation

Where a defect or system error has been identified as a major deviation, a communication must be conducted internally to initiate the implementation of this procedure.

The Project director / Site Production manager is in charge of this communication within one week of the identification of the major deviation.

Firstly, all the key information related to major deviation is gathered. (Refer to the indicative list in *Appendix 2*).

Secondly, the information is sent via email to the following address:

BYTP-Quality-Product@bouygues-construction.com

The recipients of the email (BYTP QP Manager and QE Manager) will oversee any additional distribution, if they deem necessary:

- Leader(s) of the relevant business network
- Product quality process leader.

6.3 Perform a root cause analysis

Once the major deviation is identified, a thorough investigation must be conducted to find the root cause(s) of the fault.

The Project Director / Site Production Manager is responsible for ensuring this analysis is carried out as he has the authority to mobilize the necessary human resources.

With the help of the Production Teams & Project Quality Manager, a list of individuals is established to provide insightful analysis on the sequence of events leading to the deviation. This may include BYTP employees, but also, depending on the case, representatives of a subcontractor, or possibly a co-contractor. Experts from the Technical Department of the site or headquarters (concrete, piles...) are mobilized if necessary.

The Project Quality Manager is responsible for leading and formalizing the cause analysis. If necessary, they can seek support from the Product Quality Manager.

The preferred methodology is the “A3 problem-solving” (Refer to Appendix 3 for details). Alternative methodologies can be implemented provided that the objective is to identify the root causes, i.e. the primary factor that led to the deviation. The effectiveness of corrective and preventive actions largely depends on a detailed and relevant analysis of these root causes.

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The analysis of the causes must be carried out within one month of the identification of the major deviation. This period may be extended if technical expertise is required to characterize the defect and identify its origin.

Once the root cause analysis is completed and approved by the Project Director / Site Production Manager, the Project Quality Manager will send the analysis report to the following email address: BYTP-Quality-Product@bouygues-construction.com.

Either the Product Quality Manager or QE Manager, are responsible for any additional circulation of the analysis report to the following personnel:

- Leaders (s) of the relevant business network,
- Product quality process leader

Note that the case analysis is an internal document and will remain strictly confidential. It will be used to define corrective actions at the site level and propose preventive actions that will be analysed, amended if necessary and approved during the presentation to the ExCo. (cf. §6.5).

The corrective actions will be defined during this stage, the follow-up of these actions will be under the responsibility of the Project Quality Manager.

6.4 Take ownership of root cause analysis

The Product Quality Manager is tasked with reviewing the root cause analysis in collaboration with the Project Quality Manager.

The objectives are :

- To provide an external perspective and to verify that the information provided is understandable to someone outside the project.
- To define the impacted standards that need to be updated.

The Product Quality manager, with the support of QE manager, completes the analysis report which must include a preventive action plan in relation to the identified root causes.

This stage of reviewing and defining preventive actions must be completed within a maximum of 15 days of receipt of the analysis Report.

6.5 Present the Major Deviation to ExCo

Once the root cause analysis is finalized, a presentation of Major Deviation to the ExCo is organized by the Product Quality Manager. This is to be completed within one month, following the analysis report review.

The following personnels are invited to the meeting:

- Members of ExCo
- BYTP QE Manager
- Product quality process leader

The main objective of this meeting is to:

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- **Approve preventive action plan (at BYTP level) with the aim, of avoiding the recurrence of a similar Major Deviation.**
- Decide on the means or method(s) of internal communication based on the list of impacted documents defined by the Product Quality Manager.

The circulation of information regarding the Major Deviation can be carried out in various forms:

- Flash pathology with mention of Major Deviation
- Presentation at BYTP Management Committee
- ...

This action plan (definition of the action, responsible person, objective date) is formalized as output data of the review and distributed to the participants by the Product Quality Manager.

The Major Deviation Register is kept up to date by the Product Quality Manager and they should integrate the actions decided in the meetings. It is used to calculate the KPIs of the QP process relating to major deviations.

6.6 Monitor the preventive actions for Major Deviation

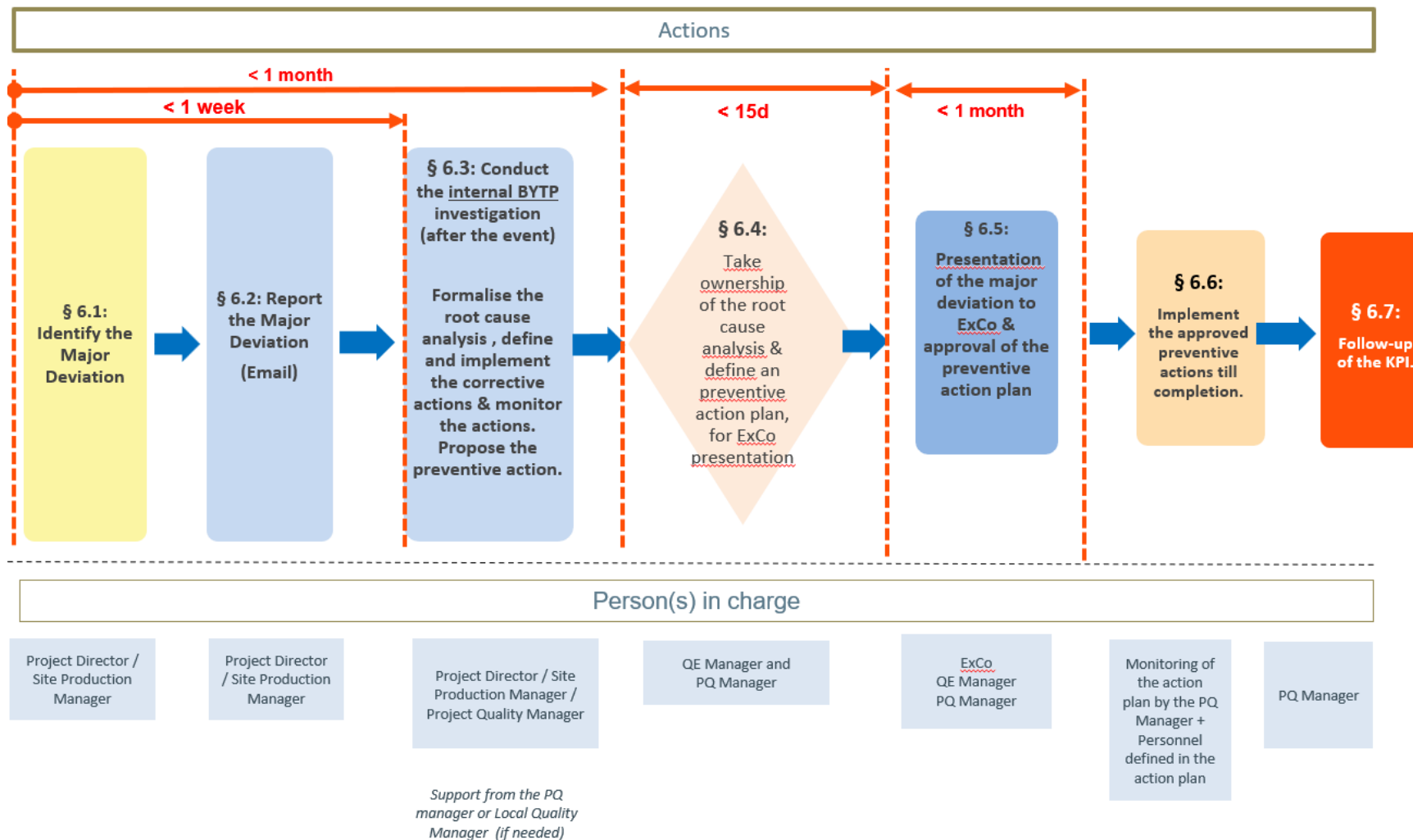
The preventive action plan is managed by the Quality Product Manager.

6.7 Follow-up of indicators

The monitoring and updating of indicators relating to Major Deviation is carried out by the Product Quality Manager:

- Number of Major Deviations
- Processing time for Major Deviations (date of initial declaration & date of presentation to ExCo)

Appendice 1 Major Deviation Management Flowchart



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Appendice 2 Appendix 2 Information to be provided for communicating a Major Deviation

- Date of occurrence
- Project
- Type of work
- Work performed directly, subcontracted or co-contracted
- Location of the deviation
- Brief description of the deviation
- Estimated cost (if known)
- Immediate or ongoing curative action

Appendice 3 A3 Problem Solving Approach

The method in brief

- Step 1: Clarify the problem
- Step 2: Break down the problem
- Step 3: Set objective
- Step 4: Analyze the root cause (using the 5 whys)
- Step 5: Define corrective actions
- Step 6: Carry out the corrective actions
- Step 7: Evaluate results and processes together
- Step 8: Standardise the process



A3 framework



A3 PROBLEM SOLVING - <Title of the problem - Name of the site - Name of the pilot> Date :

1. Identify the problem:

Context of the problem

Identification of the gap (difference between the current situation and the ideal situation)

Determination of possible preventive actions (pending resolution of the problem)

2. Breakdown of the problem into sub-problems :

Who

What

When

Where

How (Study of potential point(s) of occurrence in the process)



Summary of the sub-problem to be solved

3. Setting the SMART (Specific, Measurable, Achievable, Realistic, Timely) objective:

4. Root cause analysis:



5 & 6. Define the action plan and its deployment schedule:

Root causes	Corrective actions	Impact	Cost	Global	Resp.	Deployment schedule					

7. Define the performance evaluation system:

8. Standardise good practice: